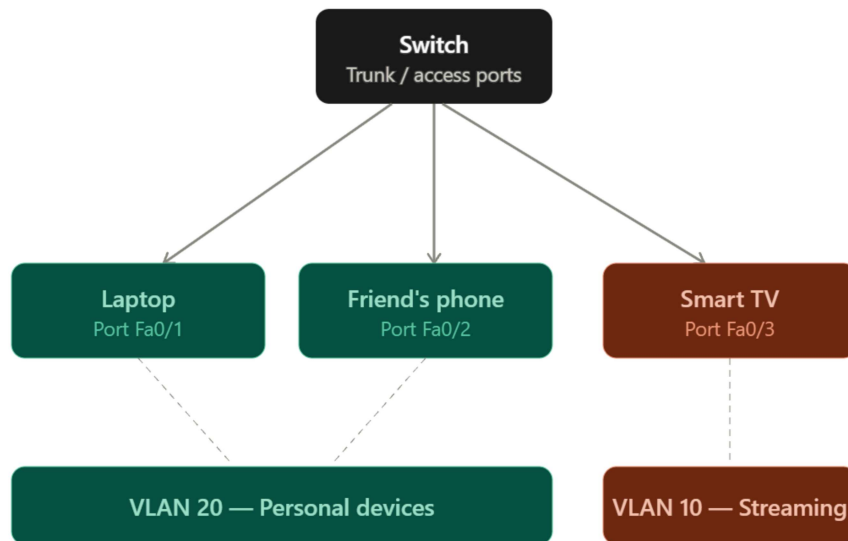


SWITCHING AND VLANS

Task 1: Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).



Task 2: Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other. Hint: Use the 'vlan' and 'switchport access vlan' commands in the simulator CLI.

File Edit Options View Tools Extensions Window Help

Physical

Logical

Root

Time: 00:18:20

PC0

Three PCs to VLAN 10 and two PCs to VLAN 20.

VLAN 10

VLAN 20

PC0

PC1

PC2

PC3

PC4

Time: 09:15:00

Simulation

Scenario 0

Fire Last Status

Cisco Packet Tracer - D:\PrctICE\Cisco packet tracer\Assesment.pkt

PC0

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.10.20:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.21

Pinging 192.168.10.21 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.10.21:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.10.7

Pinging 192.168.10.7 with 32 bytes of data:

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Reply from 192.168.10.7: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.7:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10 VLAN0010	active	Fa0/1, Fa0/2, Fa0/3
20 VLAN0020	active	Fa0/4, Fa0/5
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

Switch#Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#

TASK 3: In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

PC2

PhysicalConfigDesktopProgrammingAttributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Reply from 192.168.10.6: bytes=32 time<1ms TTL=128

Reply from 192.168.10.6: bytes=32 time=1ms TTL=128

Reply from 192.168.10.6: bytes=32 time=2ms TTL=128

Reply from 192.168.10.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.6:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Request timed out.

Request timed out.

Request timed out.

Request timed out.

Ping statistics for 192.168.10.20:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

Cisco Packet Tracer - D:\PrCTICE\Cisco packet tracer\Assessment.pkt

FileEditOptionsViewToolsExtensionsWindowHelp

LogicalPhysical

x: 635, y: 38

Root

12:02:00

Three PCs to VLAN 10 and two PCs to VLAN 20.

VLAN 10

Fa0/1

Fa0/2

Fa0/3

Fa0/4

Fa0/5

PC-PT PC0

P: 192.168.10.5

PC-PT PC1

P: 192.168.10.8

VLAN 20

Fa0

192.168.10.21

PC-PT PC4

192.168.10.20

PC-PT PC3

192.168.10.7

PC-PT PC2

The above screenshot shows the initial configuration of 3 PCs connected with VLAN 10 and 2 PCs connected with VLAN 20.

Ping from PC2 (192.168.10.7) to PC3 (192.168.10.20) and PC4 (192.168.10.21) failed

```
Switch>enable
Switch#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#switchport ?
% Unrecognized command
Switch(config)#int fa0/3
Switch(config-if)#switchport ?
    access      Set access mode characteristics of the interface
    mode        Set trunking mode of the interface
    nonegotiate  Device will not engage in negotiation protocol on this
                  interface
    port-security Security related command
    priority     Set appliance 802.1p priority
    protected    Configure an interface to be a protected port
    trunk        Set trunking characteristics of the interface
    voice        Voice appliance attributes
Switch(config-if)#switchport access ?
    vlan  Set VLAN when interface is in access mode
Switch(config-if)#switchport access vlan 20
Switch(config-if)#do wr
Building configuration...
[OK]
Switch(config-if)#exit
Switch(config)#|
```

Interface fa0/3 (PC2) is now changed from VLAN10 to VLAN20.

```
C:\>ping 192.168.10.20

Pinging 192.168.10.20 with 32 bytes of data:

Request timed out.
Reply from 192.168.10.20: bytes=32 time=3ms TTL=128
Reply from 192.168.10.20: bytes=32 time=1ms TTL=128
Reply from 192.168.10.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.20:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>ping 192.168.10.6

Pinging 192.168.10.6 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.6:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

- Ping test carried out again from PC2 to PC1 (192.168.10.6) - failed

- Ping test carried out from PC2 to PC3 (192.168.10.20) —success.

Task 4: Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3) Constraint: Use a loop or range command if supported by the tool to avoid repeating the same command 10 times.

```
Switch(config)#vlan ?
<1-4094> ISL VLAN IDs 1-1005
Switch(config)#vlan 30
Switch(config-vlan)#?
VLAN configuration commands:
  exit          Apply changes, bump revision number, and exit mode
  name          Ascii name of the VLAN
  no            Negate a command or set its defaults
  remote-span   Add the Remote Switched Port Analyzer (RSPAN) feature to the VLAN
Switch(config-vlan)#name IOTdevices
Switch(config-vlan)#do wr
Building configuration...
[OK]
Switch(config-vlan)#exit
Switch(config)#int range fa0/10-20
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 30
Switch(config-if-range)#do wr
Building configuration...
[OK]
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
```

```
Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10 VLAN0010	active	Fa0/1, Fa0/2
20 VLAN0020	active	Fa0/3, Fa0/4, Fa0/5
30 IOTdevices	active	Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
Switch#
```

